









Validating Causality



S  +

S



-

R  **+**

R



-



S



—



C



-

R  +

R.B. -



C



—





PO

PO







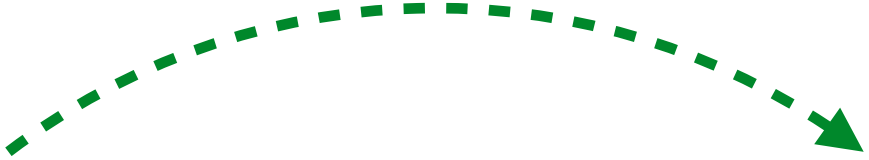
PO

po

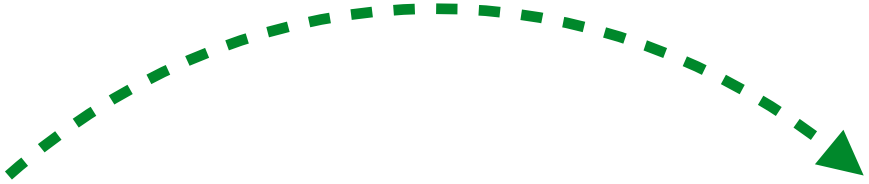


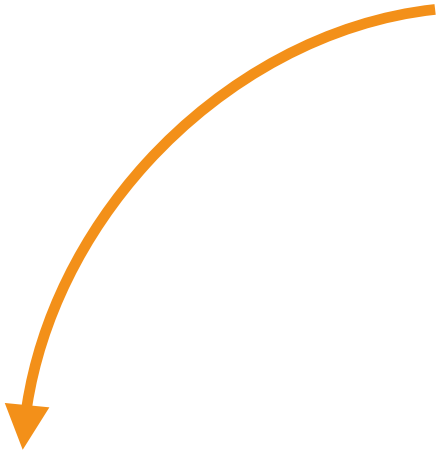
po

po

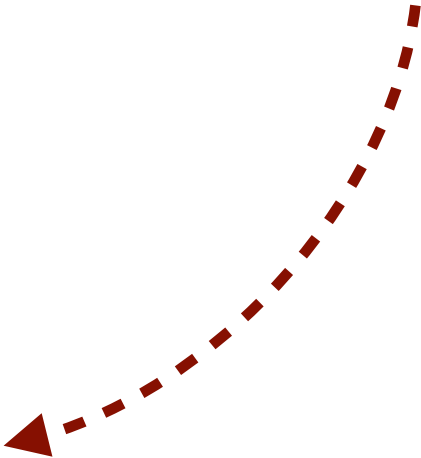








collog



h

b

```
update(src, +100)
```



```
update(src, -100)
```



```
def update(acc: cown[Account], amount: int) {  
  when(acc) {    
    acc.balance += amount  
    val id = acc.id()  
    when(log) {  log.record(id, amount) }  
  }  
}
```

R  +

R



-







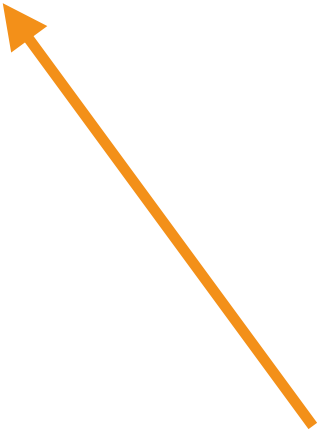
po

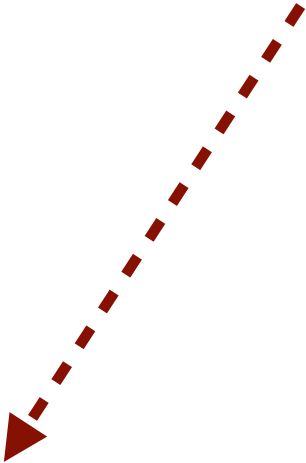
PO



collog







h

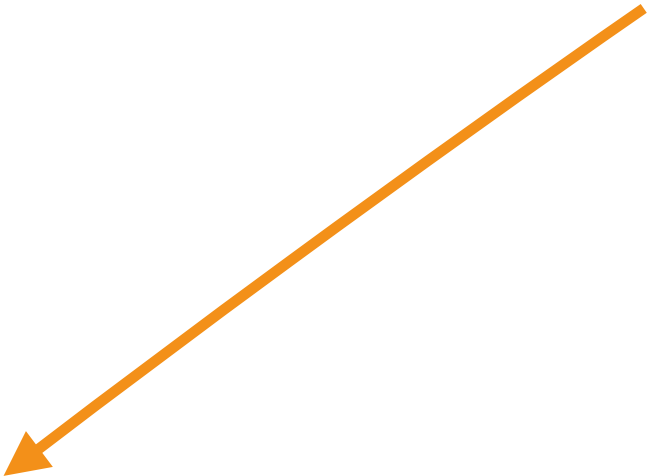
b







**Intentionally
disallowed**



COACC







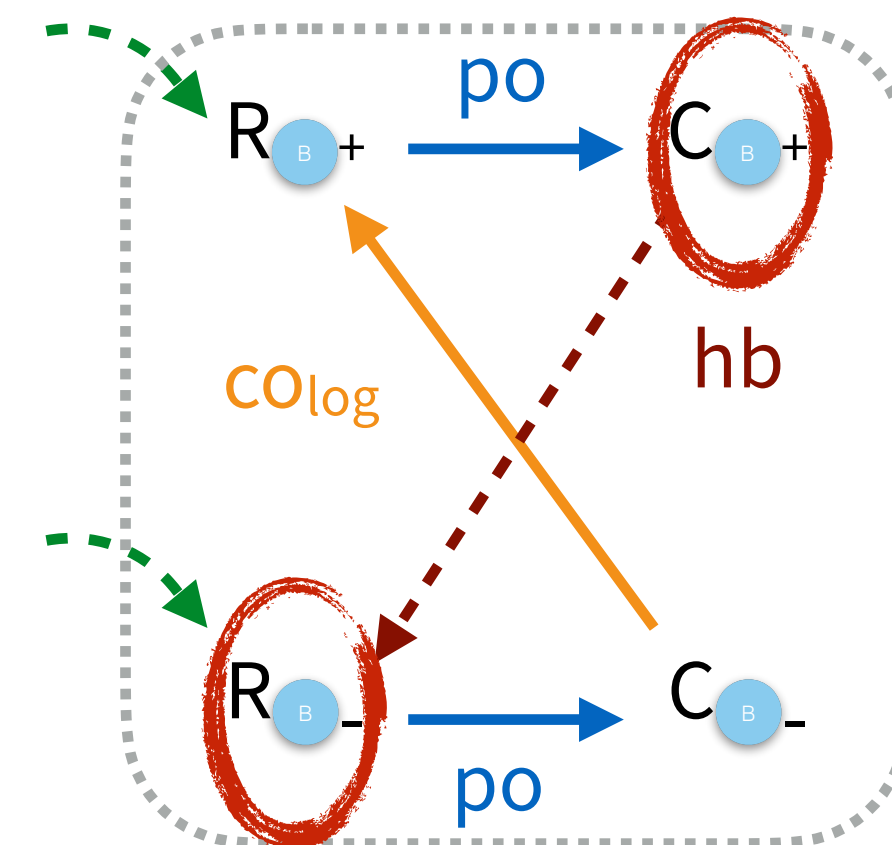
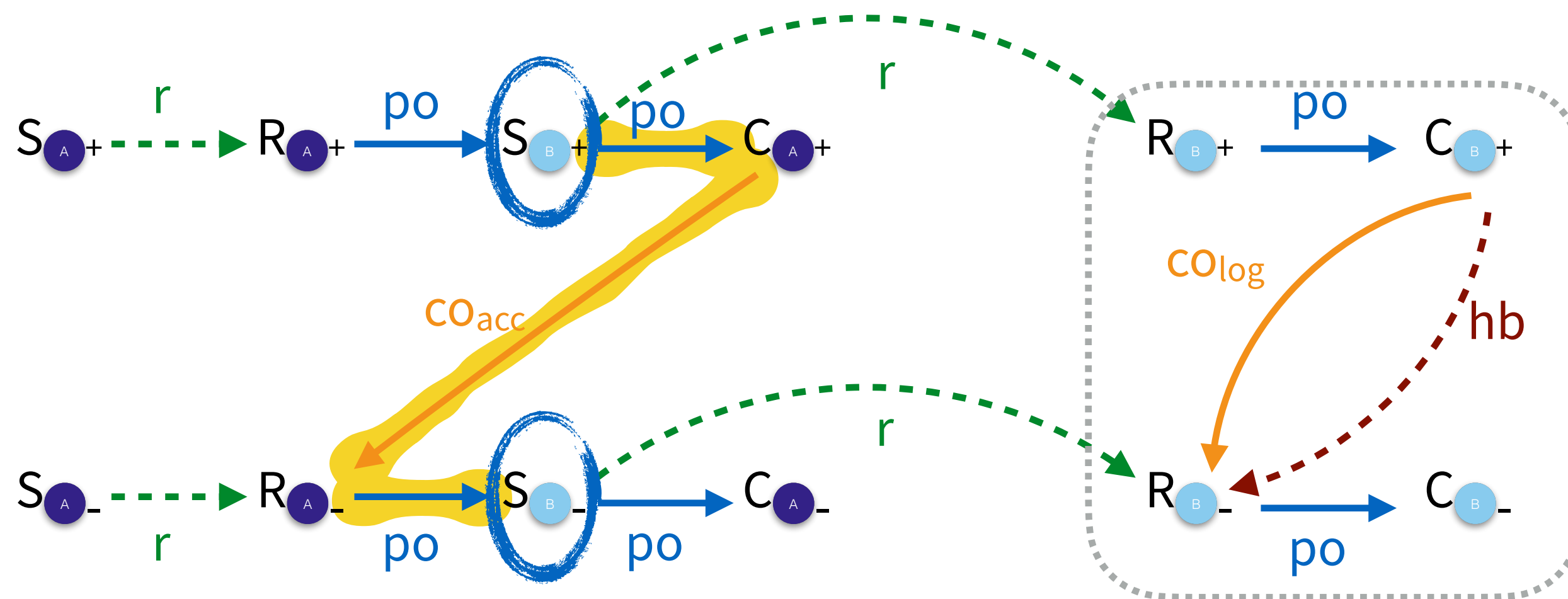


Validating Causality

```
def update(acc: cown[Account], amount: int) {  
  when(acc) { A  
    acc.balance += amount  
    val id = acc.id()  
    when(log) { B log.record(id, amount) }  
  }  
}
```

update(src, +100)

update(src, -100)



X Intentionally disallowed



What does this get us?
